

Marishka M. Mehta

Computational Researcher | LLM Evaluation & Behavioral Modeling

marishka.mehta@gmail.com · marishkamehta.github.io · LinkedIn · GitHub · Tulsa, OK · open to relocation / remote

Summary

Computational researcher and final-year PhD candidate studying decision-making in humans and large language models. Combines behavioral experimental design, Bayesian modeling, and machine learning to evaluate bias, reliability, and generalization. Builds reproducible research pipelines for LLM evaluation and behavioral, clinical, and neuroimaging data.

Experience

Graduate Research Assistant — Laureate Institute for Brain Research, Tulsa, OK Aug 2022 – present

Advisor: Dr. Ryan Smith

- Led analysis of an approach-avoidance conflict task across large, longitudinal psychiatric samples; combined frequentist and Bayesian methods with an ensemble classification pipeline that predicts diagnostic group membership out of sample.
- Published a preregistered behavioral study; built reproducible analysis pipelines for behavioral and fMRI data on SLURM-based HPC.

Clinical Research Coordinator — Depression and Anxiety Center, Icahn School of Medicine at Mount Sinai, NY Jul 2021 – Jul 2022

PI: Dr. Laurel Morris

- Designed online cognitive tasks (PsychoPy) and integrated behavioral data with REDCap and mindLAMP2 smartphone-based phenotyping.
 - Built computational and imaging analysis pipelines (MATLAB/VBA toolbox, Python, AFNI, MEICA) for 7T fMRI studies of motivation.
 - Ran IRB submissions, clinical assessments, and participant recruitment for depression and anxiety studies.
-

Technical Skills

Programming & data analysis · Python (NumPy, pandas, SciPy, Matplotlib, Jupyter), R, MATLAB

LLM evaluation · Behavioral evaluation, LLM-as-a-judge assessment, prompt ablations, evaluation task design, bias and reliability analysis, sampling-parameter sweeps

Statistics & machine learning · Bayesian hierarchical modeling (VBA), mixed-effects models, path analysis, ensemble classification (caretEnsemble), cross-validation, sensitivity analysis

Research pipelines & reproducibility · Automated scoring, dataset validation, experiment tracking, data provenance, resumable collection, Git, mock-based testing, preregistration

Model inference & research computing · vLLM, hosted model APIs, SLURM-based HPC

Cognitive modeling · Model-based/model-free reinforcement learning, active inference, DCM

Neuroimaging · 3T/7T fMRI; fMRIPrep, FSL, SPM, AFNI, MEICA, Nilearn

Experimental methods · PsychoPy, jsPsych, REDCap

Education

- **PhD, Cyber Studies** (Data Science & Psychology) - The University of Tulsa · expected Dec 2026
- **M.A., Psychology** — New York University · 2021
- **M.Sc., Neuroscience (Neurasmus Dual Diploma)** — University of Bordeaux & Laval University · 2019
- **B.Sc. (Hons.), Biomedical Science** — Queen Mary, University of London · 2015

Selected Work

- **Approach-avoidance diagnostic assessment** — predictive modeling pipeline that classifies psychiatric diagnoses out of sample using computational measures of approach-avoidance conflict. *Computational Psychiatry*, 2025. [Link](#).
- **Digital phenotyping of mental health** — smartphone-based activity and motivation measures that track depression and anxiety in the real world. *npj Digital Medicine*, 2025. [Link](#).
- **Motivation and agency as symptom-dimension markers** — computational parameters linked to anhedonia, stress, and anxiety; subjective agency over rewards reduced in depression and anxiety. *Psychological Medicine*, 2024; *PLOS ONE*, 2023.

Selected Awards

- **Dissertation Completion Fellowship** — University of Tulsa · 2026–2027
- **2nd Place, Hurricane Pitch Competition** (\$3,500) — University of Tulsa · 2024
- **Trainee Professional Development Award** (\$1,000) — Society for Neuroscience · 2025

Submitted Manuscripts

- **Mehta, M. M., & Dennis, D. K.** (2026). When Old Rules Override New Decisions: Policy-State Revision in LLM Checkers. *Submitted to the NeurIPS 2026 Workshop on Interpreting Agent Behavior*.